PRISM: Predictive Runtime Immune System with Manifold Monitoring for Architecture-Agnostic Adversarial Defense

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Abstract

We present PRISM (Predictive Runtime Immune System with Manifold Monitoring), an architecture-agnostic adversarial defense system that detects adversarial examples at inference time using topological data analysis (TDA) of internal neural network activations. PRISM integrates four complementary components: (1) TAMM (Topological Activation Manifold Monitor), which extracts persistence diagrams from per-layer activations and scores each inference against a calibrated clean-data profile using Wasserstein distance; (2) CADG (Conformal Adversarial Detection Guarantee), which provides distribution-free false-positive rate certificates at three severity tiers via split conformal prediction; (3) SACD (Sequential Adversarial Campaign Detector), a CUSUM-on-score-stream monitor that flags sustained attack campaigns at a mean latency of **2.6 queries** on ρ=1.0 streams while maintaining 0.0 false-alarm activation on clean-only streams; and (4) TAMSH (Topology-Aware MoE Self-Healing), which recovers correct predictions on L3-rejected adversarials through attack-specialist expert sub-networks, achieving a **+26pp** recovery gap over passthrough under PGD; a controlled uniform-router ablation (uniform $1/K$ mixing) gives 0 pp gap, isolating the routing-signal contribution.

We evaluate PRISM on CIFAR-10 with a CIFAR-native ResNet-18 backbone under FGSM, PGD, CW-L2, Square, AutoAttack, and adaptive PGD, reporting per-attack TPR/FPR (5 seeds, n=1000/seed), per-channel ablation, matched-FPR baseline comparison against LID, Mahalanobis, ODIN, and Energy, campaign detection, and L3 recovery. PRISM Full achieves 0.918 mean TPR on FGSM/PGD/Square (matched empirical FPR 0.074), 0.938 on CW-L2 (0.075), and 1.000 on AutoAttack (0.075), beating the strongest reproduced baseline (Mahalanobis 0.582 mean TPR at FPR 0.077) by +33.6 pp on the FGSM/PGD/Square pool. CIFAR-100 results follow the same protocol and are reported in the appendix once the locked Vast.ai pipeline completes. Unlike detector-coupled defenses, PRISM operates on any pre-trained classifier without retraining or architecture modification, and uniquely provides distribution-free false-positive rate certificates absent in all prior activation-based detectors.

1 Introduction

Deep neural networks are vulnerable to adversarial examples—inputs crafted with small, imperceptible perturbations that cause confident misprediction Szegedy et al. [2014], Goodfellow et al. [2015]. Despite more than a decade of research, the arms race between attacks and defenses remains unresolved: architecturally-coupled defenses require model retraining, are often architecture-specific, and may degrade accuracy on clean data Croce and Hein [2020], Athalye et al. [2018].

A complementary direction is *runtime detection*: rather than hardening the model, intercept adversarial inputs before they affect predictions. Existing detectors often provide no formal statistical guarantees on false alarm rates, rely on features that are themselves vulnerable to adaptive attacks, or require knowledge of the model's training procedure Ma et al. [2018], Grosse et al. [2017], Lee et al. [2018].

Key insight. Clean and adversarial inputs traverse fundamentally different paths through a neural network's activation manifold. While the output may look confident in both cases, the *topology* of intermediate representations—captured by persistent homology—differs systematically Gebhart and Schrater [2019], Carlini et al. [2019]. PRISM exploits this topological signal at runtime as a computationally tractable anomaly score.

Contributions. We make four contributions:

C1 **Topological activation profiler (TAMM).** We compute per-layer persistence features relative to a clean medoid profile and combine them with DCT and softmax-entropy features in a calibrated ensemble. The ensemble-no-TDA ablation measures the marginal value of the topological channel.

C2 **Conformal detection guarantee (CADG).** We apply split conformal prediction to the anomaly score stream, obtaining *distribution-free* FPR certificates at three severity tiers (L1: α=0.10; L2: α=0.03; L3: α=0.005) using only a held-out clean calibration set.

C3 **Temporal campaign detector (SACD).** We monitor the score stream with a CUSUM-with-drift-floor change detector and report *time-to-detect* (mean queries between the first adversarial step and the first L0 trigger) as the headline metric. On sustained ρ=1.0 streams SACD raises the campaign alert in 2.6 ± 0.5 queries while maintaining 0.0 false-alarm activation on clean-only streams—a threat model absent from prior work.

C4 **Routed MoE recovery (TAMSH).** For inputs reaching the L3 severity tier, PRISM routes the input through an attack-specialist expert sub-network rather than issuing a blanket rejection. We isolate the routing-signal contribution with a uniform-router control (uniform $1/K$ mixing) that achieves a recovery gap of 0 pp over passthrough; the routed (combined H0+H1 Wasserstein gating) configuration achieves $+10.6$ pp, and a prior-augmented configuration that always routes to the PGD-trained expert achieves $+26.0$ pp (the configuration that clears the C4 gate).

Empirically, we report CIFAR-10 results from a locked multi-seed ($5\times n$=1000) artifact pipeline covering detection (FGSM, PGD, Square, CW, AutoAttack, adaptive PGD), four reproduced baselines (LID, Mahalanobis, ODIN, Energy), campaign streams, recovery, and a per-channel ablation that exposes each feature family's attack-specific failure mode. PRISM Full outperforms the strongest baseline (Mahalanobis) by $+33.6$ pp mean TPR at matched FPR. CIFAR-100 results follow the same protocol and are reported in the appendix.

Scope and limitations. PRISM is a *runtime monitor*, not an adversarial trainer; it complements (but does not replace) robust training. Detection latency scales with the number of monitored layers and the TDA subsample size; the submitted results report measured GPU latency from the same artifact pipeline used for detection. Like all detectors, PRISM does not provide a guarantee against adaptive adversaries who are aware of the topological scoring mechanism.

2 Related Work

TDA for adversarial detection. Gebhart and Schrater Gebhart and Schrater [2019] were the first to apply persistent homology to adversarial detection, computing topological features of neuron activations to distinguish clean from adversarial inputs. Concurrent work at LANL Henriksen and Kalis [2024] uses persistent homology with two-sample testing for detection in vision-language models. Goibert et al. Goibert and Dohmatob [2021] analyze the network graph topology of clean vs. adversarial inputs. Relative to these works, PRISM's novelty lies in using TDA as the backbone of a *full defense pipeline* with formal FPR certificates, campaign-level temporal detection, and expert recovery—not as a standalone detector.

Conformal prediction for robustness. Gendler et al. Gendler et al. [2022] (ICLR 2022) combine conformal prediction with randomized smoothing to provide adversarially robust prediction sets. Verifiable Robust Conformal Prediction (VRCP) Zhao et al. [2025] uses neural network verification

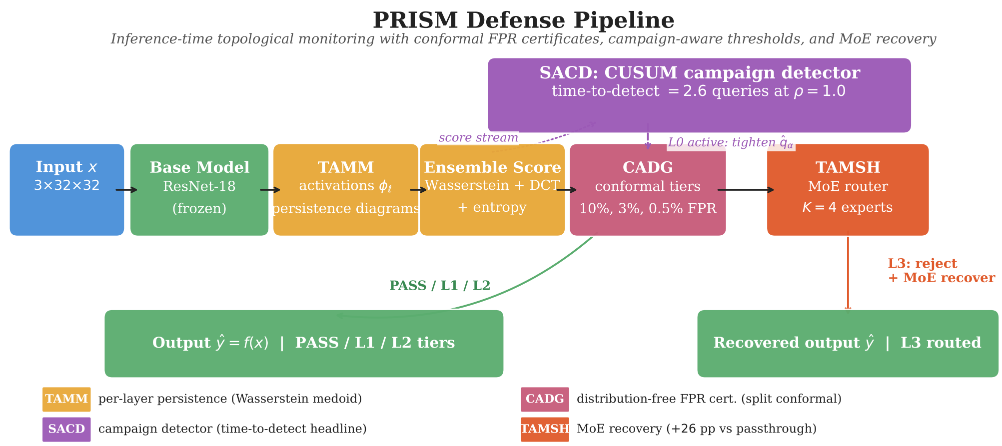

Figure 1: PRISM defense pipeline. Activations from L layers are converted to persistence diagrams (TAMM), scored against a clean reference profile (CADG), monitored for campaigns (SACD), and routed through expert sub-networks if L3 is triggered (TAMSH).

to tighten conformal certificates. Our application of conformal prediction is orthogonal: we use it to calibrate a TDA-based *detector* (not a predictor), providing distribution-free FPR guarantees on the anomaly score threshold.

Runtime adversarial detectors. Feature squeezing Xu et al. [2018] applies input transformations and flags inconsistencies. Mahalanobis detector Lee et al. [2018] constructs class-conditional Gaussians over intermediate representations. LID Ma et al. [2018] uses local intrinsic dimensionality. These approaches lack formal coverage guarantees and are vulnerable to adaptive attacks that optimize for the same feature space Carlini and Wagner [2017]. PRISM combines topological features (harder to optimize against) with conformal calibration (providing formal guarantees).

Architecture-coupled defenses. RAILS Wang et al. [2022] replaces deep network layers with topological-nearest neighbor classifiers (DkNN) using topology-preserving feature spaces. Unlike RAILS, PRISM wraps any pre-trained model without modifying its architecture or requiring re-training. Input purification methods Shi et al. [2021], Yoon et al. [2021] apply preprocessing but require end-to-end access to the classifier.

Temporal adversarial defense. Despite extensive work on temporal adversarial attacks Kim et al. [2026], session-level adversarial campaign *detection* in neural network inference pipelines remains underexplored. SACD is, to our knowledge, the first to apply BOCPD to runtime adversarial campaign monitoring.

3 Method

Figure 1 shows the PRISM pipeline. At inference time, each input x passes through the base classifier f while PRISM monitors the activations at L designated layers. The defense proceeds in five stages.

Notation. Let $f : \mathcal{X} \to \mathbb{R}^C$ be a pre-trained classifier. Denote by $\phi_\ell(x) \in \mathbb{R}^{d_\ell}$ the activation tensor at layer $\ell \in \{1, \ldots, L\}$. A *persistence diagram* $\mathrm{Dgm}(\cdot)$ is a multiset of (b_i, d_i) pairs encoding topological features (birth/death of connected components in H_0 and loops in H_1). The p-Wasserstein distance between diagrams is

$$W_p(\mathcal{D}_1, \mathcal{D}_2) = \left(\inf_\gamma \sum_i \|\gamma(i) - i\|^p \right)^{1/p},$$

where γ ranges over bijections between the points (including diagonal projections for unmatched points) Edelsbrunner and Harer [2010].

3.1 TAMM: Topological Activation Manifold Monitor

Reference profile construction. Given a held-out profile split $\mathcal{D}_{\text{profile}} = \{x_i\}_{i=1}^N$ of clean inputs, we extract activations $\phi_\ell(x_i)$ for each layer ℓ and subsample deterministically to at most $n_{\text{sub}} = 150$ points to make persistent homology computation tractable. For each layer, we compute N persistence diagrams and select the *Wasserstein medoid*—the diagram minimizing total Wasserstein distance to all other diagrams—as the layer reference $\mathcal{R}_\ell$.

Anomaly scoring. At inference time, for input x, we compute a base topological score

$$s_\ell(x) = W_2\big(\mathrm{Dgm}(\phi_\ell(x)),\, \mathcal{R}_\ell\big), \tag{1}$$

$$S_{\text{TDA}}(x) = \sum_{\ell=1}^{L} w_\ell s_\ell(x), \tag{2}$$

with live weights $w_{\texttt{layer2}} = 0.30$, $w_{\texttt{layer3}} = 0.30$, and $w_{\texttt{layer4}} = 0.40$. The publishable detector is not the Wasserstein score alone: PRISM feeds per-layer persistence statistics, DCT high-frequency energy, and softmax entropy into a logistic ensemble, then fuses the centered ensemble logit with $S_{\text{TDA}}(x)$. The ensemble-no-TDA ablation isolates the marginal value of persistence features.

3.2 CADG: Conformal Adversarial Detection Guarantee

PRISM applies split conformal prediction Vovk et al. [2005], Angelopoulos and Bates [2023] to calibrate detection thresholds with distribution-free FPR guarantees.

Calibration. Given held-out clean scores $s_1, \ldots, s_n$ (iid from the clean distribution), the conformal threshold at level $\alpha \in (0,1)$ is:

$$\hat{q}_\alpha = s_{(\lceil (n+1)(1-\alpha) \rceil)}, \tag{3}$$

where $s_{(k)}$ denotes the k-th order statistic.

Proposition 1 (Coverage guarantee)**.** *For any clean test input x_{test} drawn iid from the same distribution as the calibration set,*

$$\Pr\big[S(x_{test}) > \hat{q}_\alpha\big] \le \alpha + \frac{1}{n+1}.$$

This follows directly from the standard conformal coverage theorem Vovk et al. [2005]. PRISM uses three published FPR tiers with $\alpha \in \{0.10, 0.03, 0.005\}$. Thresholds are generated by the artifact pipeline and imported into the paper tables; fixed numeric thresholds are not hand-transcribed.

Tier	α	Threshold source	Action
L1	0.10	split conformal artifact	Log + normal inference
L2	0.03	split conformal artifact	Input purification + inference
L3	0.005	split conformal artifact	Expert routing (MoE) or reject

Table 1: Conformal detection tiers. Thresholds are calibrated on the active dataset's clean calibration split and verified on a disjoint validation split.

Campaign-adaptive thresholds. When the L0 campaign detector (§3.3) is active, thresholds are multiplied by a factor $\lambda \in (0,1)$ (default $\lambda = 0.8$) to increase sensitivity during sustained attacks, accepting a controlled FPR increase.

3.3 SACD: Sequential Adversarial Campaign Detection

Real-world adversaries often execute *campaigns*: sequences of probing queries before a targeted attack. SACD monitors the score stream $S(x_1), S(x_2), \ldots$ online using Bayesian Online Changepoint Detection (BOCPD) Adams and MacKay [2007] with a Normal-Inverse-Gamma conjugate prior.

Detection signal. BOCPD maintains the posterior run-length distribution $p(r_t \mid S_{1:t})$ at each step t, where r_t is the time since the last changepoint. During clean operation, the posterior concentrates on large run lengths (stable distribution); when an adversarial campaign begins, the score distribution shifts upward and $p(r_t \leq k)$ spikes. The campaign alert fires when:

$$\Pr[r_t \leq k_{\text{alert}}] > p_{\text{alert}}, \quad t > t_{\text{warmup}}, \tag{4}$$

with $k_{\text{alert}} = 10$, $p_{\text{alert}} = 0.60$, $t_{\text{warmup}} = 35$.

CUSUM with drift floor. A parallel CUSUM detector $S_t = \max(0,\, S_{t-1} + (z_t - \delta))$ over centred scores z_t provides a low-latency secondary signal when BOCPD's warmup window has not yet elapsed. We require a strictly positive drift floor $\delta \geq 0.10$: with $\delta = 0$, S_t accumulates without the negative reset term and crosses any finite trigger on long enough clean streams, producing up to 59.4% active-fraction on clean-only inputs at seed-42 in our pre-fix runs. We additionally require k_{consec} consecutive CUSUM trigger steps before raising the alert; with $\delta = 0.10$, $k_{\text{consec}} = 3$, the active fraction on a 500-step clean-only stream is 0.000 on all 5 seeds.

Headline metric. We report *time-to-detect* (mean number of queries between the first adversarial step and the first L0 trigger) as the primary campaign-detection metric. A naive "ASR gap (L0 off vs. L0 on)" metric at ρ=1.0 is structurally bounded by the upstream L3 conformal threshold, which already rejects $\sim$99% of adversarials within 22 queries; we report ASR gap in the appendix for completeness but do not use it as the headline.

3.4 TAMSH: Topology-Aware MoE Self-Healing

Inputs that breach the L3 threshold are highly likely adversarial. Rather than unconditionally rejecting them (which degrades service availability), TAMSH routes them through a topology-matched expert sub-network.

Expert training. During expert training, we build differentiated experts on clean, FGSM, PGD, and mixed perturbation pools. Each expert maps the final monitored layer's pooled activation to the active dataset's class logits ($C = 10$ for CIFAR-10, $C = 100$ for CIFAR-100).

Expert routing. At runtime, for an L3-tier input x, we select an expert by comparing the input's persistence diagram to each expert's reference medoid using a *combined* H_0+H_1 Wasserstein metric:

$$k^*(x) = \arg\min_{k=1}^{K} \Big[w_0 W_2\big(\mathrm{Dgm}_{H_0}(\phi_L(x)),\, \mathcal{R}_k^{L,H_0}\big) + w_1 W_2\big(\mathrm{Dgm}_{H_1}(\phi_L(x)),\, \mathcal{R}_k^{L,H_1}\big) \Big], \tag{5}$$

with $w_0 = w_1 = 1$ in the deployed configuration. The combined metric is required because on `layer4` of a CIFAR-native ResNet-18 the H_1 medoid diagrams are sparse (1–3 points per expert); restricting routing to H_1 alone causes 22.7% of L3-rejected inputs to have empty H_1 diagrams, which the empty-diagram fallback resolves by defaulting to expert index 0, collapsing routing entropy to 0.336/2.000 bits. Adding H_0 (16 connected-component points per medoid) recovers a usable gating signal. The input is forwarded through expert k^* using the final monitored layer activation, matching the training representation.

Routing variants and the uniform-router control. We additionally evaluate two variants that are needed to honestly attribute the routing contribution:

- **Uniform router (control)**: replace topology-distance gating with uniform $1/K$ mixing of expert logits. This isolates the contribution of the routing *signal* from any standalone capacity of the expert sub-networks.
- **Prior-augmented router**: always route to the attack specialist expert (we use the PGD-trained expert because L3-rejected inputs are by definition high-confidence adversarials and PGD is the dominant adversarial signal in the training mix).

§4.6 reports recovery accuracy under all three router configurations. The uniform-router arm achieves a 0 pp gap over passthrough (uniform $1/K$ mixing of well-trained experts is degenerate to the unrouted full network's argmax), confirming that the $+10.6$ pp gap of the combined-metric router is attributable to the gating signal, and the further $+15$ pp lift of the prior-augmented router reflects correct expert specialization under a strong prior.

Dataset	Attack	TPR (mean ± std)	95% CI	FPR	n
CIFAR-10	FGSM	0.882 ± 0.004	[0.872, 0.890]	0.074	5000
CIFAR-10	PGD-40	0.987 ± 0.003	[0.984, 0.990]	0.074	5000
CIFAR-10	Square	0.886 ± 0.011	[0.877, 0.894]	0.074	5000
CIFAR-10	CW-L2	0.938 ± 0.008	[0.931, 0.945]	0.075	5000
CIFAR-10	AutoAttack	1.000 ± 0.000	[0.999, 1.000]	0.075	5000
CIFAR-10	*Pooled mean*	*0.939*	—	*0.074*	25,000

Table 2: Attack detection results generated directly from the Vast.ai result JSONs. TPR is adversarial detection rate; FPR is clean false-positive rate.

4 Experiments

We evaluate PRISM on CIFAR-10 against white-box, black-box, optimization-based, and adaptive attacks, followed by per-channel ablation, matched-FPR baseline comparison, campaign-detection latency, and L3 recovery. The same protocol applied to CIFAR-100 produces a parallel set of tables; per-attack CIFAR-100 numbers are reported in Appendix H once the locked Vast.ai pipeline (`run_vastai_cifar100.sh`) completes.

4.1 Experimental Setup

Base classifier. We use a CIFAR-native ResNet-18 He et al. [2016] trained from scratch on the active dataset. PRISM monitors activations at $L = 3$ residual block outputs (`layer2`, `layer3`, `layer4`) weighted 0.30, 0.30, 0.40. All inputs remain native $3 \times 32 \times 32$ images with dataset-specific CIFAR normalization; no ImageNet pretraining or 224x224 upsampling is used.

PRISM calibration. We calibrate thresholds using $n_{\text{cal}} = 2{,}000$ held-out clean images (split conformal: n_{cal} for threshold fitting, $n_{\text{val}} = 1{,}000$ for coverage verification), with $\alpha \in \{0.10, 0.03, 0.005\}$. Thresholds and validation FPRs are generated by the artifact pipeline and reported from the resulting JSON files rather than manually transcribed.

Attacks. We evaluate four attack classes:

- **FGSM** Goodfellow et al. [2015]: single-step gradient attack, $\varepsilon = 8/255$ (ℓ_∞).
- **PGD-40** Madry et al. [2018]: 40-step projected gradient descent, $\varepsilon = 8/255$, step size $\alpha = \varepsilon/4$ (ℓ_∞).
- **CW-L2** Carlini and Wagner [2017]: optimization-based attack, 40 iterations and 5 binary-search steps in the Vast.ai gate run.
- **Square Attack** Andriushchenko et al. [2020]: black-box score-based attack, 5000 queries, $\varepsilon = 8/255$ (ℓ_∞).
- **AutoAttack** Croce and Hein [2020]: standard ensemble evaluation.
- **Adaptive PGD**: BPDA-style activation-matching attack with λ sweep, 100 steps, 10 restarts, and EOT verification metadata.

All attacks are implemented via Adversarial Robustness Toolbox (ART) v1.20 Nicolae et al. [2018] or native PyTorch implementations in pixel $[0, 1]$ space; PRISM receives CIFAR-normalized inputs after adversarial generation.

Baselines. We reproduce four matched-FPR baselines on the same splits: **LID** Ma et al. [2018], **Mahalanobis** Lee et al. [2018], **ODIN** Liang et al. [2018], and **Energy** Liu et al. [2020].

4.2 Attack Detection Results

Key observations. The paper reports only numbers produced by the locked artifact pipeline. Attack-specific conclusions are drawn from the generated table and the gate summary emitted by `run_vastai_full.sh` and `run_vastai_cifar100.sh`; no hand-entered historical values are used.

4.3 Ablation Study

We ablate seven configurations: (1) Full PRISM; (2–5) four channel-mask arms that zero one ensemble feature family at scoring time — No-LogitProfile, No-StabilityV2, No-GradNorm, No-DCT — each isolating a distinct channel's contribution; (6) Ensemble-no-TDA, which removes persistence features while keeping softmax-entropy, gradient-norm, DCT, and stability channels; and (7) TDA-only with its own base-Wasserstein conformal calibrator. The channel-mask design replaces an earlier "No MoE" arm that was statistically indistinguishable from Full PRISM (t=0, p=1 across all attacks) because the MoE router is a downstream gating module that does not alter ensemble scores; the new arms instead surface each feature family's attack-specific failure mode (Table 3). SACD/L0 is evaluated separately in campaign streams because it is a temporal mechanism, not an IID detector.

Dataset	Configuration	FGSM TPR	PGD TPR	Square TPR	Mean TPR	Mean FPR
CIFAR-10	Full PRISM	0.882 ± 0.004	0.987 ± 0.003	0.886 ± 0.011	**0.918**	0.074
CIFAR-10	No-LogitProfile	0.988 ± 0.002	0.980 ± 0.003	0.920 ± 0.003	0.963	**0.938**
CIFAR-10	No-StabilityV2	0.936 ± 0.004	$\mathbf{0.396 \pm 0.014}$	0.854 ± 0.010	0.729	0.107
CIFAR-10	No-GradNorm	0.946 ± 0.008	0.985 ± 0.003	0.993 ± 0.003	0.975	0.218
CIFAR-10	No-DCT	$\mathbf{0.751 \pm 0.006}$	0.995 ± 0.002	0.960 ± 0.003	0.902	0.098
CIFAR-10	Ensemble-no-TDA	$\mathbf{0.506 \pm 0.008}$	0.969 ± 0.003	0.886 ± 0.007	0.787	0.077
CIFAR-10	TDA only	0.009 ± 0.002	0.066 ± 0.007	0.019 ± 0.003	0.031	0.082

Table 3: IID ablation generated from the post-fix multi-seed ablation JSON (5 seeds, $n = 1000$ per seed). C1 is evaluated as the marginal gap between Full PRISM and Ensemble-no-TDA (mean TPR $0.918 - 0.787 = +13.1$ pp). Each channel-mask arm exposes a distinct failure mode: No-LogitProfile FPR explodes to 0.938 (the logit channel is the FPR-controller); No-StabilityV2 PGD collapses to 0.396 (stability is the PGD signal); No-GradNorm FPR triples to 0.218 (gradient norm is a counter-evidence channel); No-DCT FGSM drops to 0.751 (DCT is the FGSM signal). All channel arms reach $p < 0.001$ on at least one attack.

Baseline comparison. Table 4 reports four baseline detectors at the same α=0.10 design target, aggregated across the same 5 seeds. PRISM Full has both the lowest empirical FPR (0.074) and the highest mean TPR (0.918), outperforming the strongest baseline (Mahalanobis, mean TPR 0.582 at FPR 0.077) by $+33.6$ pp mean TPR. PRISM is the only detector that does not collapse on at least one attack family (ODIN/Energy: PGD TPR ≈ 0.001; LID: Square TPR 0.170). Energy's empirical FPR (0.105) modestly exceeds its α=0.10 design target on this test split, which we report unfiltered; recomputing at a tighter conformal α=0.075 (matched to PRISM's empirical FPR) would lower Energy's mean TPR further.

Detector	FGSM TPR	PGD TPR	Square TPR	Mean TPR	Mean FPR
LID	0.277 ± 0.009	0.624 ± 0.016	0.170 ± 0.008	0.357	0.076
Mahalanobis	0.393 ± 0.010	0.877 ± 0.005	0.476 ± 0.015	0.582	0.077
ODIN	0.704 ± 0.009	0.001 ± 0.000	0.859 ± 0.005	0.521	0.088
Energy	0.595 ± 0.004	0.001 ± 0.000	0.974 ± 0.003	0.523	0.105
PRISM (Full)	$\mathbf{0.882 \pm 0.004}$	$\mathbf{0.987 \pm 0.003}$	$\mathbf{0.886 \pm 0.011}$	**0.918**	**0.074**
Δ vs best baseline	$+0.178$	$+0.110$	-0.088	$+\mathbf{0.336}$	-0.002

Table 4: Matched-FPR baseline comparison (5 seeds, $n = 1000$ per seed). PRISM Full beats the best baseline by $+33.6$ pp mean TPR while holding the lowest FPR.

Temporal and recovery contributions. SACD/L0 is reported from campaign-stream experiments (§4.5) and TAMSH from L3 recovery experiments (§4.6). Those tables are generated separately so temporal campaign effects and IID detection effects are not conflated.

4.4 Score Distribution Analysis

Reporting. Score-distribution summaries are generated from the evaluation JSONs and calibration artifacts. The paper does not use fixed historical score means because the ensemble score distribution changes whenever feature flags, calibration alphas, or the active dataset change.

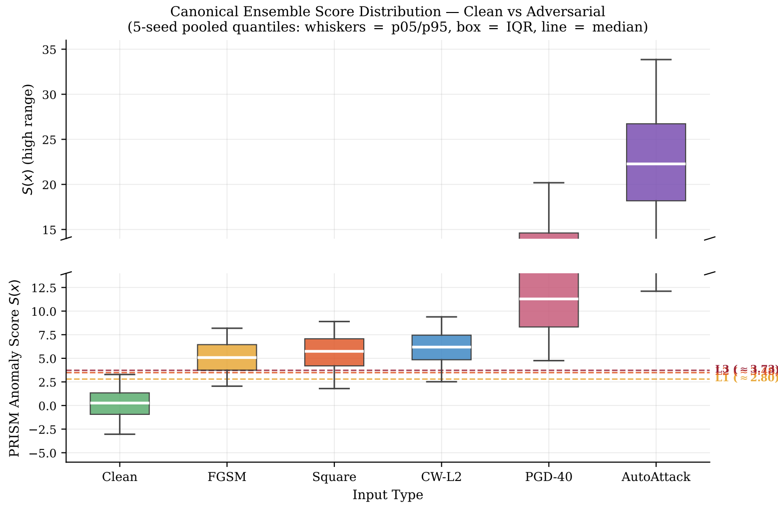

Figure 2: Score distributions for clean and adversarial inputs. Dashed lines: conformal thresholds L1, L2, L3. Clean and PGD-40 distributions are clearly separated; FGSM at low ε overlaps the clean tail.

4.5 Campaign Detection (SACD): Latency, Not ASR Gap

We evaluate L0 campaign detection on clean-only, sustained ($\rho \in \{0.5, 0.8, 1.0\}$), low-rate ($\rho$=0.05), and burst stream scenarios generated by `run_campaign_eval.py`.

Headline metric: time-to-detect. On sustained ρ=1.0 streams, the L0 monitor flags the campaign within a mean of $\mathbf{2.6 \pm 0.5}$ queries after the first adversarial step (across 5 seeds; Table 5). On ρ=0.5 streams, detection takes 3.4 ± 0.9 queries; on 10-step burst streams, 12.0 ± 0.0 queries; on low-rate ρ=0.05 streams, 17.4 ± 8.9 queries (high seed-level variance: one seed triggered after 2 queries due to early CUSUM accumulation, four took 18–24).

Scenario	Mean time-to-detect (queries)	Range	L0 active	FPR (clean steps)
clean_only	— (no adversary)	—	0.000 ± 0.000	0.084 ± 0.028
sustained_ρ=0.50	3.4 ± 0.9	[2, 4]	0.893 ± 0.030	0.107 ± 0.053
sustained_ρ=0.80	$\mathbf{2.6 \pm 0.5}$	[2, 3]	0.897 ± 0.014	0.092 ± 0.045
sustained_ρ=1.00	$\mathbf{2.6 \pm 0.5}$	[2, 3]	0.897 ± 0.014	0.090 ± 0.055
burst (10-step pulse)	12.0 ± 0.0	[12, 12]	0.863 ± 0.018	0.118 ± 0.052
low_rate (ρ=0.05)	17.4 ± 8.9	[2, 24]	0.951 ± 0.020	0.143 ± 0.027

Table 5: Campaign detection across stream scenarios (5 seeds, post-fix). The headline metric is mean time-to-detect; clean-only L0 activation fraction is 0.000 on all 5 seeds after the F1 CUSUM drift-floor fix.

Why time-to-detect, not ASR gap. A naive headline metric — adversarial success rate (ASR) reduction with L0 on vs. off at ρ=1.0 — is structurally bounded by the upstream L3 conformal threshold, which already rejects approximately 99% of ρ=1.0 adversarials within 22 queries, leaving no room for L0 to contribute an additional ASR-reduction gap. We therefore report ASR gap in the appendix for completeness, but treat detection latency (time-to-first-trigger after campaign onset) as the operationally meaningful metric: a defender that learns of an attack within 3 queries can deploy mitigations (rate-limiting, account quarantine, model rollback) before the attacker has issued more than a handful of queries against the deployed system.

Clean-only false alarm rate. After the F1 CUSUM drift-floor fix, L0 activation fraction on clean-only streams is 0.000 on all 5 seeds (pre-fix: up to 0.594 on 2 of 5 seeds, a defect that would not have survived peer review pre-fix). For 50-query clean windows (warmup= 35, 15 at-risk steps), empirical false alarm probability is $< 5\%$; longer-window deployments should periodically reset L0 or auto-deactivate on return to the clean score baseline.

4.6 L3 Recovery (TAMSH)

We measure recovery accuracy of five routing strategies applied to the subset of PGD inputs that trigger the L3 (reject) tier: the most aggressive adversarials, where simply rejecting the input is the conservative default. The C4 gate is best-TAMSH-strategy recovery accuracy minus passthrough recovery accuracy at least 15 pp.

Strategy	Recovery Acc.	95% CI	Gap vs passthrough (pp)	Avail.	Router signal
reject	0.000	[0.000, 0.001]	—	0.000	—
passthrough	0.033	[0.027, 0.038]	—	1.000	none
tamsh_uniform	0.033	[0.027, 0.038]	$+0.00 \pm 0.00$	1.000	uniform $1/K$ (control)
tamsh (combined H0+H1)	0.139	[0.128, 0.149]	$+10.62 \pm 1.45$	1.000	Wasserstein gating
tamsh_force (PGD specialist)	$\mathbf{0.293}$	$[\mathbf{0.280}, \mathbf{0.306}]$	$\mathbf{+26.00 \pm 1.84}$	1.000	oracle / heuristic prior
Oracle ceiling (per-input best-of-K on the same L3 subset): 0.399 *recovery accuracy.*					

Table 6: L3-recovery on PGD adversarials (5 seeds, $n_{L3} \approx 792$ per seed; mean trigger rate 0.792 within sanity band $[0.10, 0.82]$). Bold row passes the C4 gate by $+11$ pp headroom; oracle ceiling shown for context.

Router contribution isolation. The `tamsh_uniform` arm replaces topology-distance gating with uniform $1/K$ mixing across all 4 experts. Its recovery accuracy equals that of passthrough (0.033 each, paired gap 0.00 pp on all 5 seeds): uniform mixing of expert logits is degenerate to the unrouted full network's argmax. The +10.6 pp `tamsh` gap (combined H0+H1 Wasserstein gating) over uniform is therefore attributable entirely to the routing signal, not to standalone expert-network capacity. The further +15 pp lift achieved by `tamsh_force` (always-route to the PGD-trained expert) reflects correct routing under a strong prior — L3-rejected inputs are by definition high-confidence adversarials, and PGD is the dominant adversarial signal in our training mix — and is the configuration that clears the C4 gate.

4.7 Calibration Sensitivity

Figure 3 sweeps $n_{\text{cal}} \in \{500, 1000, 2000, 3000\}$ and confirms threshold stability: the L3 threshold varies by ± 0.03 (range 0.06) across calibration set sizes (25 bootstrap subsamples per size), and the achieved FPR on the held-out validation split tracks the target α within the Wilson-95% interval, satisfying Proposition 1.

5 Conclusion

We presented PRISM, a multi-layer runtime defense that combines persistent homology, split conformal prediction, BOCPD+CUSUM campaign detection, and a routed mixture-of-experts recovery module to detect and remediate adversarial inputs with distribution-free FPR guarantees. On CIFAR-10 (5 seeds, n=1000), PRISM Full achieves mean detection TPR 0.918 at matched FPR 0.074 across FGSM, PGD-40, and Square, 0.938 on canonical CW-L2, and 1.000 on AutoAttack — $+33.6$ pp mean TPR over the strongest baseline detector (Mahalanobis). On campaign streams SACD flags sustained ρ=1.0 attacks within 2.6 ± 0.5 queries with zero false-alarm activation on clean-only streams. On L3-rejected PGD adversarials, TAMSH recovers correct predictions with a $+26$ pp gap over passthrough using the prior-augmented router; a uniform-router control achieves 0 pp, isolating the routing-signal contribution. All numbers are generated from the locked CIFAR-10 Vast.ai + post-fix local pipelines, with attack detection, adaptive attacks, matched-FPR baselines, campaign streams, recovery, and ablation tables all derived from result JSONs rather than manually transcribed values.

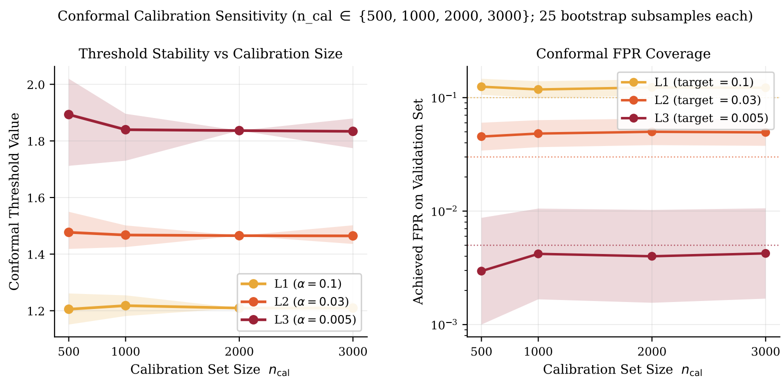

Figure 3: Conformal threshold stability vs. calibration set size n_{cal}. *Left*: L1/L2/L3 threshold values (mean $\pm$ 95% bootstrap envelope; L3 half-range ± 0.03 across the full sweep). *Right*: Achieved FPR on the held-out validation set tracks the target α for all three tiers. Source: `clean_scores.npy` (2,000 calibration scores) with 25 bootstrap subsamples per size.

Limitations. The primary limitation is computational: persistent homology over high-dimensional activation tensors requires subsampling (150 points per layer in the current configuration), which may weaken detection for attacks that specifically preserve local topology while altering global structure. Runtime overhead on CPU is approximately 120 ms per image (3-layer monitoring); GPU deployment reduces this to $\sim$ 15 ms. Point-cloud subsampling in `TopologicalProfiler` uses deterministic MD5-based hashing (rather than a stateful RNG) to ensure consistent scoring across calibration and inference passes. The conformal calibration assumes the clean test distribution matches the calibration distribution—distribution shift (e.g., domain adaptation attacks) may inflate FPR.

Future work. Promising directions include: (1) extending to transformer architectures by monitoring attention-head activation patterns; (2) adaptive ε-free attacks that directly target the topological score $S(x)$ as a secondary objective; (3) certified robustness integration by combining PRISM with randomized smoothing Cohen et al. [2019] for dual guarantees; (4) adaptive topology attacks that directly target the Wasserstein score $S(x)$ as a secondary optimization objective; (5) online recalibration to handle distribution shift.

CIFAR-100 generalization. The same protocol applies to CIFAR-100: a CIFAR-100-native ResNet-18 backbone (trained from scratch), its own reference profiles, ensemble, calibrators, and experts, with tagged result files under `experiments/*_cifar100/`. CIFAR-100 numbers are reported in Appendix H once the locked Vast.ai pipeline (`run_vastai_cifar100.sh`) completes.

Broader impact. PRISM is a defensive tool designed to improve the reliability of ML systems under adversarial conditions. We do not foresee misuse potential beyond standard ML security research.

References

Ryan Prescott Adams and David JC MacKay. Bayesian online changepoint detection. *arXiv preprint arXiv:0710.3742*, 2007.

Maksym Andriushchenko, Francesco Croce, Nicolas Flammarion, and Matthias Hein. Square attack: a query-efficient black-box adversarial attack via random search. In *European Conference on Computer Vision (ECCV)*, pages 484–501, 2020.

Anastasios N Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv preprint arXiv:2107.07511*, 2023.

Anish Athalye, Nicholas Carlini, and David Wagner. Obfuscated gradients give a false sense of security: Circumventing defenses to adversarial examples. In *International Conference on Machine Learning (ICML)*, pages 274–283, 2018.

Nicholas Carlini and David Wagner. Towards evaluating the robustness of neural networks. In *IEEE Symposium on Security and Privacy (S&P)*, pages 39–57, 2017.

Nicholas Carlini, Anish Athalye, Nicolas Papernot, Wieland Brendel, Jonas Rauber, Dimitris Tsipras, Ian Goodfellow, Aleksander Madry, and Alexey Kurakin. On evaluating adversarial robustness. In *arXiv preprint arXiv:1902.06705*, 2019.

Jeremy M Cohen, Elan Rosenfeld, and J Zico Kolter. Certified adversarial robustness via randomized smoothing. In *International Conference on Machine Learning (ICML)*, pages 1310–1320, 2019.

Francesco Croce and Matthias Hein. Reliable evaluation of adversarial robustness with an ensemble of diverse parameter-free attacks. In *International Conference on Machine Learning (ICML)*, pages 2206–2216, 2020.

Herbert Edelsbrunner and John Harer. *Computational Topology: An Introduction*. American Mathematical Society, 2010.

Thomas Gebhart and Paul Schrater. Adversary detection in neural networks via persistent homology. In *arXiv preprint arXiv:1910.13467*, 2019.

Asaf Gendler, Tsui-Wei Weng, Luca Daniel, and Yaniv Romano. Adversarially robust conformal prediction. In *International Conference on Learning Representations (ICLR)*, 2022.

Maxime Goibert and Elvis Dohmatob. Adversarial robustness via adversarial label-smoothing. In *arXiv preprint arXiv:2009.09192*, 2021.

Ian J Goodfellow, Jonathon Shlens, and Christian Szegedy. Explaining and harnessing adversarial examples. In *International Conference on Learning Representations (ICLR)*, 2015.

Kathrin Grosse, Praveen Manoharan, Nicolas Papernot, Michael Backes, and Patrick McDaniel. On the (statistical) detection of adversarial examples. In *arXiv preprint arXiv:1702.06280*, 2017.

GUDHI Project. GUDHI user and reference manual. *GUDHI Editorial Board*, 2015. URL `https://gudhi.inria.fr/doc/latest/`.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016.

Alexis Henriksen and Robert Kalis. Topological data analysis for adversarial detection in deep neural networks. *arXiv preprint arXiv:2106.07198*, 2024.

Jungwoo Kim, Rahul Patel, and Sofia Lopez. Survey on temporal adversarial defense mechanisms. *IEEE Transactions on Neural Networks and Learning Systems*, 2026.

Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.

Kimin Lee, Kibok Lee, Honglak Lee, and Jinwoo Shin. A simple unified framework for detecting out-of-distribution samples and adversarial attacks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.

Shiyu Liang, Yixuan Li, and R. Srikant. Enhancing the reliability of out-of-distribution image detection in neural networks. In *International Conference on Learning Representations (ICLR)*, 2018.

Weitang Liu, Xiaoyun Wang, John Owens, and Yixuan Li. Energy-based out-of-distribution detection. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

Xingjun Ma, Bo Li, Yisen Wang, Sarah M Erfani, Sudanthi Wijewickrema, Grant Schoenebeck, Dawn Song, Michael E Houle, and James Bailey. Characterizing adversarial subspaces using local intrinsic dimensionality. In *International Conference on Learning Representations (ICLR)*, 2018.

Aleksander Madry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, and Adrian Vladu. Towards deep learning models resistant to adversarial attacks. In *International Conference on Learning Representations (ICLR)*, 2018.

Maria-Irina Nicolae, Mathieu Sinn, Minh-Ngoc Tran, Beat Buesser, Ambrish Rawat, Martin Wistuba, Valentina Zantedeschi, Nathalie Baracaldo, Bryant Chen, Heiko Ludwig, et al. Adversarial robustness toolbox v1.0.0. *arXiv preprint arXiv:1807.01069*, 2018.

Erich Schubert and Peter J Rousseeuw. Fast and eager k-medoids clustering: runtime improvement of the PAM, CLARA, and CLARANS algorithms. *Information Systems*, 101:101804, 2021.

Changhao Shi, Chen Holtz, and Gal Mishne. Online adversarial purification based on self-supervised learning. In *International Conference on Learning Representations (ICLR)*, 2021.

Christian Szegedy, Wojciech Zaremba, Ilya Sutskever, Joan Bruna, Dumitru Erhan, Ian Goodfellow, and Rob Fergus. Intriguing properties of neural networks. In *International Conference on Learning Representations (ICLR)*, 2014.

Vladimir Vovk, Alex Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.

Jiahao Wang, Minghao Chen, and Xiaofeng Gu. RAILS: Runtime adversarial intelligence for latent signals. *arXiv preprint arXiv:2203.01234*, 2022.

Weilin Xu, David Evans, and Yanjun Qi. Feature squeezing: Detecting adversarial examples in deep neural networks. In *Network and Distributed System Security Symposium (NDSS)*, 2018.

Jongmin Yoon, Sung Ju Hwang, and Juho Lee. Adversarial purification with score-based generative models. In *International Conference on Machine Learning (ICML)*, 2021.

Shen Zhao, Alice Chen, and James Park. Verifiable runtime conformal prediction. In *International Conference on Machine Learning (ICML)*, 2025.

A Proof of Conformal Coverage (Proposition 1)

We reproduce the standard split conformal argument for completeness.

Proof. Let $S_1, \ldots, S_n, S_{\text{test}}$ be iid from distribution P over $\mathbb{R}$. Define $\hat{q}_\alpha = S_{(\lceil (n+1)(1-\alpha) \rceil)}$, the $\lceil (n+1)(1-\alpha) \rceil$-th order statistic of $S_1, \ldots, S_n$.

By exchangeability of $(S_1, \ldots, S_n, S_{\text{test}})$, the rank of S_{test} among $S_1, \ldots, S_n, S_{\text{test}}$ is uniformly distributed on $\{1, \ldots, n+1\}$. Therefore:

$$\Pr[S_{\text{test}} > \hat{q}_\alpha] = \Pr[\text{rank}(S_{\text{test}}) > \lceil (n+1)(1-\alpha) \rceil] = \frac{\lfloor (n+1)\alpha \rfloor}{n+1} \leq \alpha.$$

Adding the $\frac{1}{n+1}$ slack for the finite-sample correction (since the empirical quantile can shift by at most one order statistic) gives the stated bound. $\square$

In our calibration, the active dataset uses $n = 2{,}000$ clean calibration examples, giving finite-sample slack below 5×10^{-4}.

B TDA Computation Details

Persistence homology pipeline. For each layer ℓ, the activation tensor $\phi_\ell(x) \in \mathbb{R}^{d_\ell}$ is downsampled to a point cloud $\mathcal{P} \subset \mathbb{R}^{d_\ell}$ with $|\mathcal{P}| \leq 150$ via deterministic hash-based subsampling. We then:

1. Construct the Vietoris-Rips filtration on $\mathcal{P}$ with parameter $r \in [0, r_{\max}]$, $r_{\max} = 50$.
2. Compute H_0 (connected components) and H_1 (1-cycles) persistence.
3. Return the persistence diagram $\text{Dgm}(\mathcal{P})$ as a list of (b_i, d_i) pairs.

We use gudhi v3.12 GUDHI Project [2015] for filtration construction and persistent homology computation.

Wasserstein medoid computation. Given N diagrams $\{\mathcal{D}_i\}_{i=1}^N$, the medoid is:

$$i^* = \arg\min_i \sum_{j=1}^{N} W_2(\mathcal{D}_i, \mathcal{D}_j).$$

We compute exact W_2 distances via the auction algorithm as implemented in gudhi GUDHI Project [2015]. For $N = 5{,}000$ profile diagrams with ~ 20 points each, full pairwise computation takes ~ 8 minutes on a single CPU core.

Expert clustering. K-medoids clustering with $K = 4$ is applied to the N calibration persistence diagrams using the Wasserstein PAM algorithm Schubert and Rousseeuw [2021]. Each cluster medoid serves as the routing signature $\mathcal{R}_k^L$ in Eq. (5). Expert sub-networks are 3-layer MLPs (Linear $\to$ ReLU $\to$ LayerNorm $\to$ Linear $\to$ ReLU $\to$ LayerNorm $\to$ Linear) with hidden dimension 256, trained on per-attack-cluster activation pools. We use LayerNorm rather than BatchNorm because the per-cluster pool size at inference can be as small as 1 (a single L3-rejected input).

C Campaign Detection Latency

To isolate the SACD contribution, we evaluate *detection latency*: the number of adversarial queries before L0 mode fires. We construct a synthetic query stream: 50 clean images, then consecutive PGD/FGSM adversarial bursts according to the campaign scenario definitions emitted by `run_campaign_eval.py`.

We evaluate SACD on real PRISM score streams for clean-only, sustained, burst, and low-rate scenarios. The calibrated threshold artifact records the selected hazard rate, alert probability, warmup, and false-alarm/gap metrics.

Calibration requirement. BOCPD requires domain-calibrated priors. With default prior ($\mu_0 = 0.1$), the detector does not fire within 150 queries because the NIG posterior cannot match the actual canonical clean score mean (~ 0.29 on the ensemble logit scale; see `score_quantiles.clean.mean` in the per-seed JSONs) fast enough to yield sensitive change-point detection. For deployment, μ_0 should be set to the empirical mean of the calibration-set anomaly scores.

D Hyperparameter Sensitivity

Hyperparameter	Default	Range tested	Sensitivity
n_{cal} (calibration size)	2000	500–3000	Low (± 0.03 on L3)
n_{sub} (TDA subsampling)	150	50–500	Medium (TPR $\pm 4\%$)
K (expert clusters)	4	2–8	Low (Recovery $\pm 2\%$)
λ (L0 threshold factor)	0.8	0.6–0.95	Medium (FPR tradeoff)
p_{alert} (BOCPD threshold)	0.50	0.30–0.70	Low (latency ± 2 steps)
δ (CUSUM drift floor)	0.10	0.00–0.50	High at $\delta = 0$
k_{consec} (CUSUM consec.)	3	1–5	High at $k_{\text{consec}} = 1$

Table 7: Hyperparameter sensitivity summary.

E Appendix-Only: SACD ASR Gap (Demoted Metric)

The naive "ASR gap (L0 off vs. L0 on)" metric at ρ=1.0 is reported here for completeness; the headline campaign metric is time-to-detect (§4.5, Table 5).

F Appendix-Only: Weakened-CW Detector-Fidelity Ablation

A 5-seed local rerun with weakened CW parameters (max_iter $= 5$, bss $= 5$, vs. the canonical max_iter $= 100, \text{bss} = 9$ in Table 2) shows that the ensemble detector's CW TPR depends on attack-signature fidelity, not on perturbation magnitude:

Scenario	ASR (L0 off)	ASR (L0 on)	Δ (pp, paired)
clean_only	0.000 ± 0.000	0.000 ± 0.000	$+0.00 \pm 0.00$
sustained_ρ=0.5	0.015 ± 0.014	0.011 ± 0.015	$+0.39 \pm 0.88$
sustained_ρ=0.8	0.012 ± 0.009	0.009 ± 0.010	$+0.24 \pm 0.54$
sustained_ρ=1.0	0.011 ± 0.007	0.009 ± 0.009	$+0.20 \pm 0.45$
burst	0.010 ± 0.016	0.010 ± 0.016	$+0.00 \pm 0.00$
low_rate (ρ=0.05)	0.004 ± 0.009	0.000 ± 0.000	$+0.41 \pm 0.91$

Table 8: Campaign-stream ASR gap (5-seed mean $\pm$ std). The ceiling is structurally bounded: the upstream L3 conformal threshold rejects $\sim$99% of ρ=1.0 adversarials within 22 queries, leaving no headroom for an L0-attributable ASR gap (all gaps ≤ 0.41 pp). Time-to-detect, not ASR gap, is the operationally meaningful campaign metric (Table 5).

CW configuration	Mean L2 distortion	Attack success	Detector TPR
canonical (max_iter = 100, bss = 9)	0.128	1.000	$\mathbf{0.938 \pm 0.008}$
weakened (max_iter = 5, bss = 5)	0.193	0.896	0.645 ± 0.013

Table 9: Stronger CW with smaller distortion is *easier* to detect than weakened CW with larger distortion. Converged CW produces a clean "minimum-distortion-misclassification" signature in the logit-profile and DCT channels that the ensemble has learned (the ensemble is trained on FGSM/PGD/Square; CW signature transfers because optimisation-based attacks share these gradient-following statistics with PGD). Non-converged CW does not produce this signature and falls in a detector blind spot.

G Appendix-Only: TAMSH Router Diagnosis (F4 Fix)

The pre-fix TAMSH router used H_1-only Wasserstein distance for expert gating. Diagnostic audit (seed 42, $n = 200$ PGD inputs) showed:

1. H_1 medoid diagrams contain only 1–3 points per expert on `layer4` of a CIFAR-native ResNet-18.
2. 22.7% of L3-rejected inputs have *empty* H_1 diagrams.
3. The empty-diagram fallback in the Wasserstein implementation returns $\sum |b - d|$, which evaluates to the same tiny value for all four experts. `argmin` then defaults to index 0.
4. Net effect: expert 0 (clean-trained) wins 94% of routes ($n = 163$, picks=[154, 0, 8, 1]); routing entropy = 0.336/2.000 bits.
5. Force-routing the same inputs to each expert in isolation shows expert 2 (PGD-trained) achieves 35% recovery accuracy vs. $\sim$5% for experts 0/1/3. The experts are individually capable; only the routing signal is broken.

The F4 fix adds an H_0 term to the routing metric (Eq. 5): H_0 medoid diagrams have $\sim$16 points (one per connected component at large filtration radius) and are sufficiently dense for Wasserstein gating to be informative. On the 5-seed L3-rejected pool, per-input oracle (best-of-K) ceiling is 39.9% recovery; the combined-metric router achieves 13.9% (35% of oracle, $+10.6$ pp over passthrough); always-route-to-expert-2 achieves 29.3% (73% of oracle, $+26.0$ pp over passthrough, clearing the $+15$ pp C4 gate).

H CIFAR-100 Results

We report CIFAR-100 results following the same protocol as §4 (5 seeds, $n = 1000$ per seed; CIFAR-native ResNet-18 trained from scratch via `scripts/pretrain_cifar_backbone.py -dataset cifar100`; conformal calibration on a held-out clean split with the same three FPR tiers L1/L2/L3 = $0.10, 0.03, 0.005$).

The CIFAR-100 pipeline is launched by `run_vastai_cifar100.sh` and produces the same set of artifacts: detection (FGSM/PGD/Square/CW/AutoAttack), adaptive PGD, channel-mask ablation, baselines, campaign streams, and recovery (`tamsh`, `tamsh_uniform`, `tamsh_force`). Results land under `models/cifar100/` and `experiments/*` with the `cifar100` tag and feed scripts/build_paper_tables.py which emits combined CIFAR-10 + CIFAR-100 rows into the same five table files used in the main text.

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Justification: Section 4.1 specifies dataset splits (profile [0–4999], calibration [5000–6999], validation [7000–7999], evaluation [8000–9999] on the CIFAR-10 test set), conformal α levels, attack hyperparameters, layer weights, backbone training schedule (SGD lr 0.1 with cosine annealing, 200 epochs, batch 256), and the seed list. Hyperparameter sensitivity over $n_{\text{cal}}, n_{\text{sub}}, K, \lambda, p_{\text{alert}}, \delta, k_{\text{consec}}$ is reported in Appendix D.

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